SimHawNet: A Modified Hawkes Process for Temporal Network Simulation

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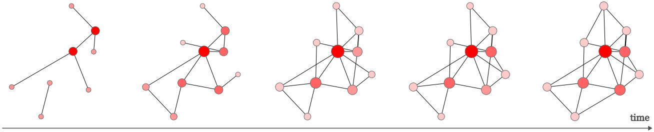
Context and Motivation



Why temporal networks?

Capture time-stamped interactions, not just static links.

Applications: social network, exchanges of emails...

Limitations of most existing models

Rely on discrete-time snapshots.

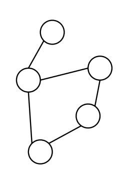
Fail to capture causal effects and temporal pattern.

Our goal

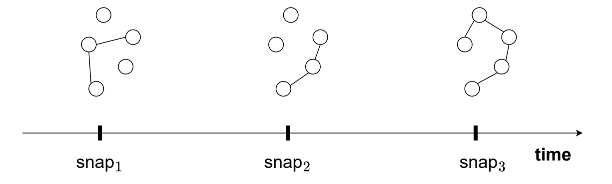
A generative framework for *continuous-time* temporal networks.

Context: From Static to Continuous-Time Networks

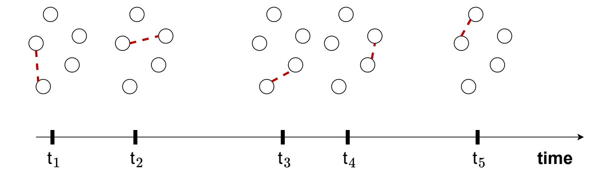
Static Graph ✗



Discrete Snapshots !

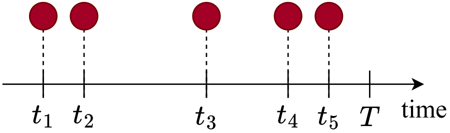


Continuous-Time ✓



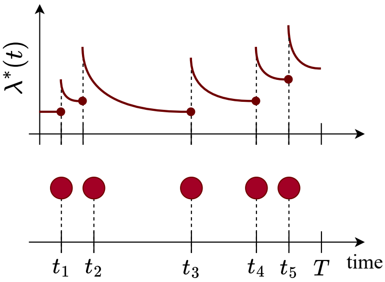
Our goal: *Model when interactions happen, not just if they happen.*

Hawkes Process : Temporal Point Process



Temporal point processes : A stochastic process whose realization is a collection of points, with each event having a well-defined position t_j in continuous time.

Hawkes Process : Intensity Function



Conditional intensity function of a Hawkes process, with μ a base intensity:

$$\lambda(t \mid \mathcal{H}) = \mu + \sum_{t_i \in \mathcal{H}_t} h(t - t_i)$$

- Dependent on the past $\mathcal{H}_t = \{t_i \in \mathcal{H} \mid t_i < t\}$
- Self-exciting: past events increase future likelihood

Problem Statement

Temporal Network:

Source and destination nodes $\mathcal{U}, \mathcal{V}$, edges $\mathcal{E} \subset \mathcal{U} \times \mathcal{V}$.

Network events : $(e_m, t_m)_{m=1}^{M}$, with $e_m = (u_m, v_m) \in \mathcal{E}$

Event history : $\mathcal{H}_t = \{(e_m, t_m) \in \mathcal{H} \mid t_m < t\}$, $\quad \mathcal{H}^{(e)} = \{(e_m, t_m) \in \mathcal{H} \mid e_m = e\}$.

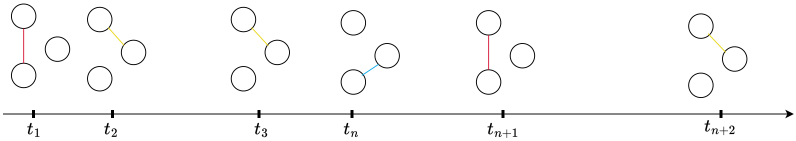
Dynamic Graph Simulator:

Observes past events $\mathcal{H}_{T_o}$ and generates future events $\mathcal{H}_{\geq T_o}$.

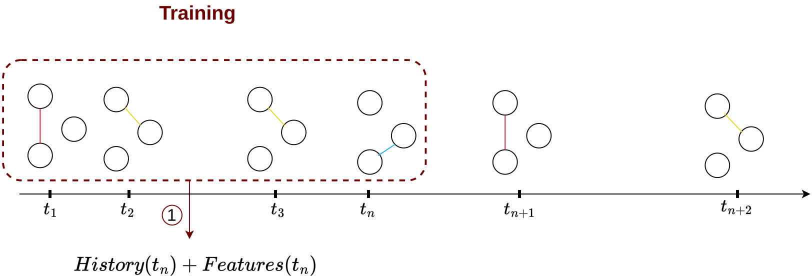
Properties: Topological Correctness and Temporal Consistency.

Temporal point processes provide a flexible modeling framework.

SimHawNet : Dataset

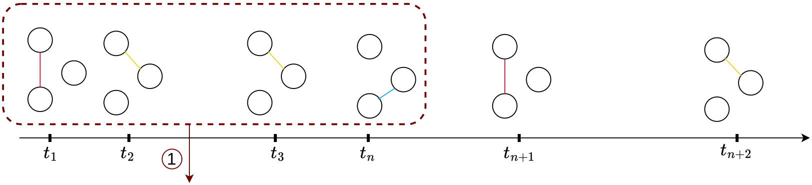


SimHawNet : Extraction



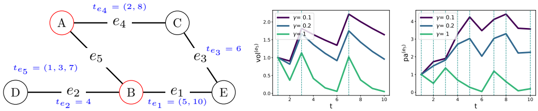
SimHawNet : Extraction

Training

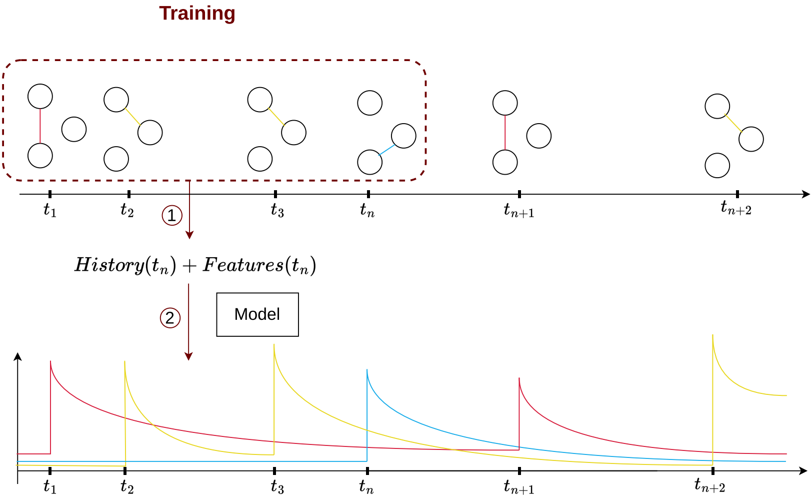


$$History(t_n) + Features(t_n)$$

1 **Extraction** of the temporal graph's history and hand-crafted creation of features



SimHawNet : Training



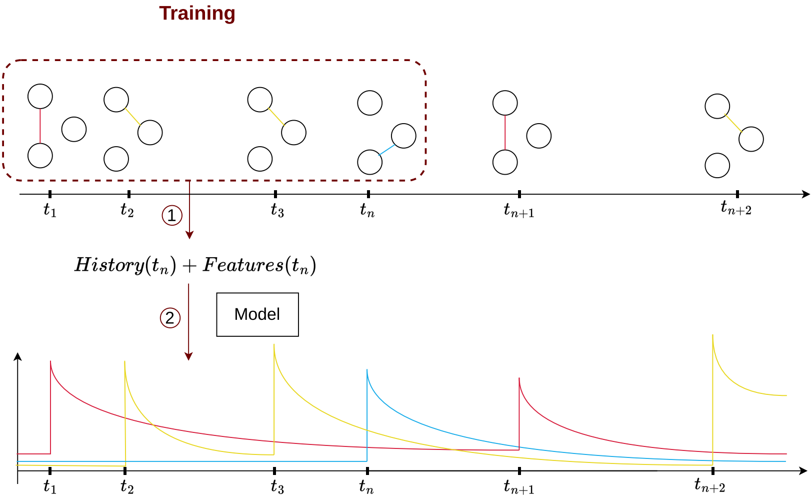
SimHawNet : Training

② **Training** the model to convert the graph into edge-wise intensity

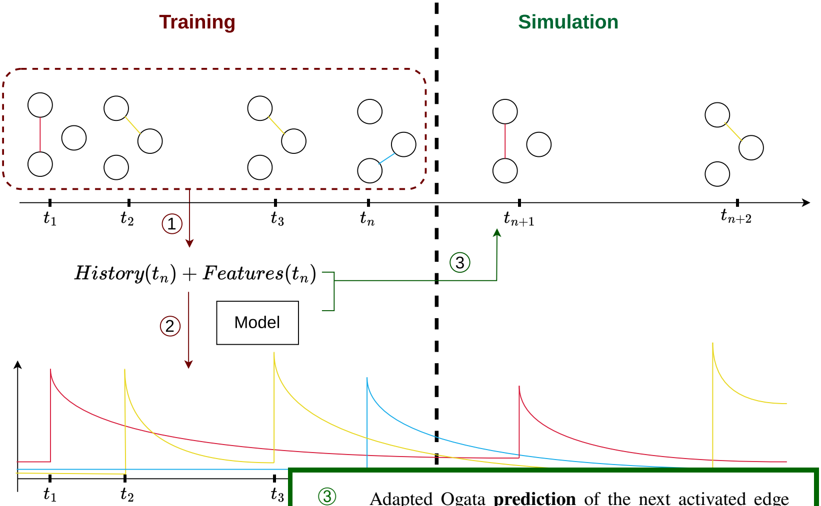
$$\underbrace{\lambda^{(e)}(t \mid \mathcal{H}_t^{(e)})}_{\text{Edge intensity}} = \overbrace{\mu^{(e)}}^{\text{Static part}} + \overbrace{\alpha^{(e)}(t)e^{-\beta^{(e)}(t-t_{last}^{(e)})}}^{\text{Dynamic part}}$$

$$\alpha^{(e)}(t) = \sigma\left(\overbrace{g}^{\text{Aggregation function}}(\underbrace{F^{(e)}(t)}_{\text{Temporal features}})\right)$$

SimHawNet : Prediction

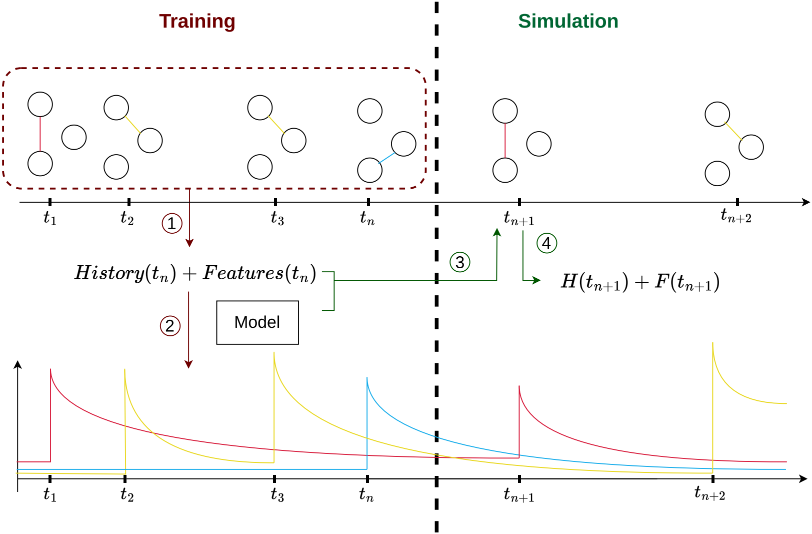


SimHawNet : Prediction

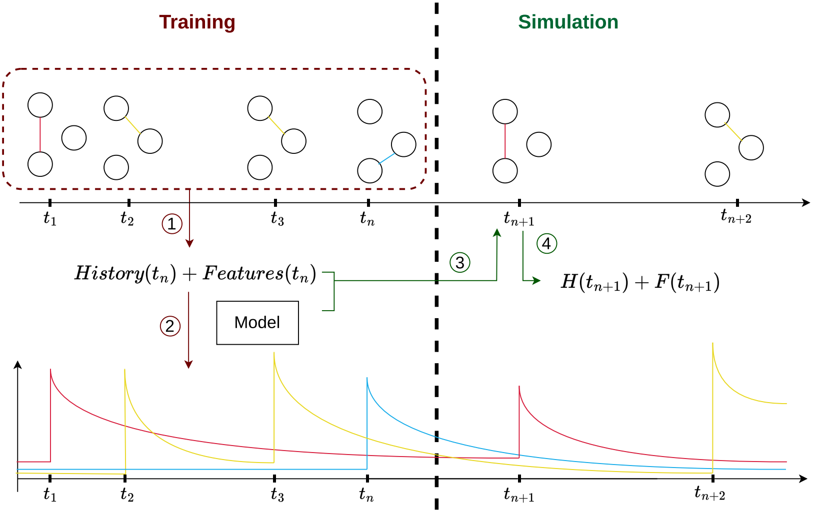


③ Adapted Ogata **prediction** of the next activated edge and the corresponding time based on the trained model, graph history and features

SimHawNet : Updating

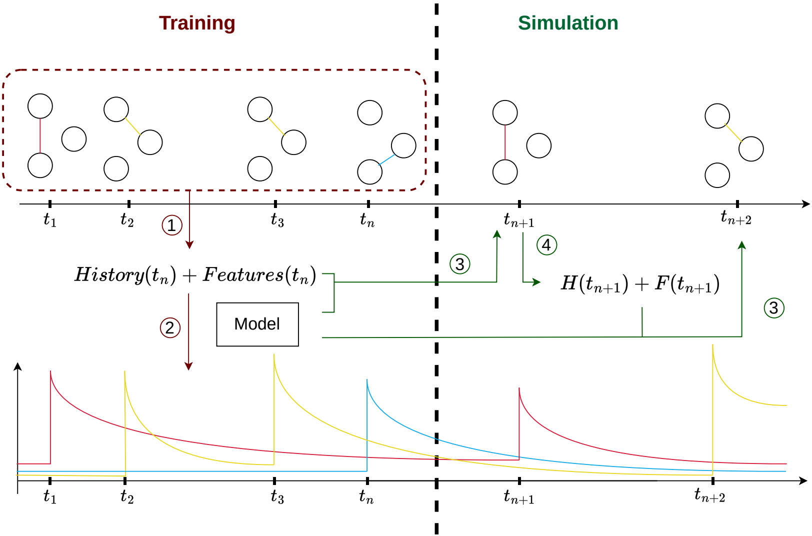


SimHawNet : Updating

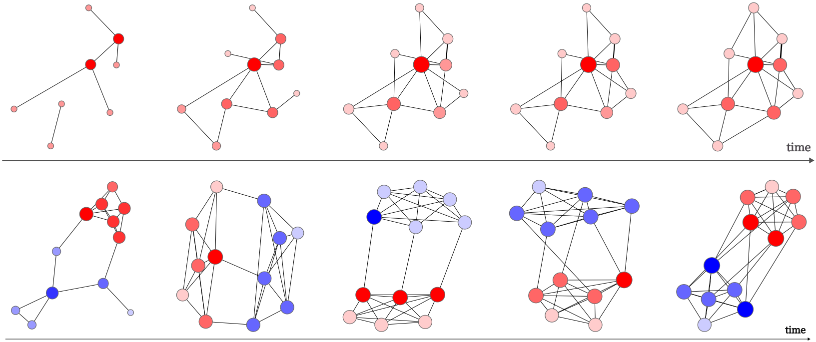


4 **Updating** the graph's history and features.

SimHawNet : Simulation



Dataset

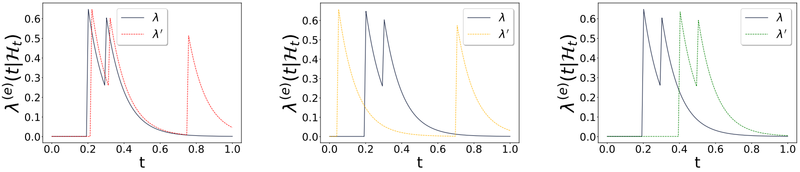


Temporal snapshots of synthetic dynamic graphs generated from the **top row:** config graph, and **bottom row:** SBM.

Metrics

→ **Topological Correctness** : Clustering Coefficient focusing on the local (sub)structures of the network and number of events.

→ **Temporal Consistency** : Dynamic Time Warping



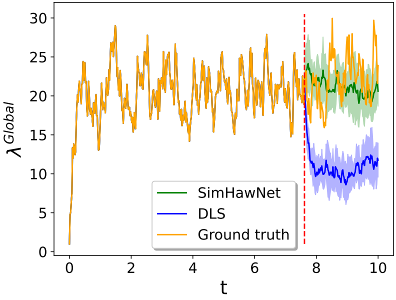
(a) $DTW = 0.91$ (b) $DTW = 1.17$ (c) $DTW = 0.04$

Results

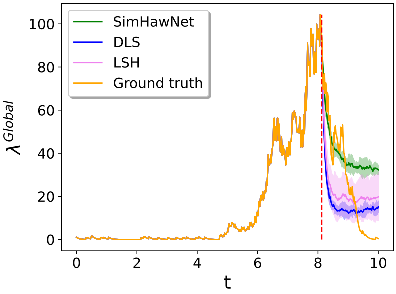
Network	Approach	# events	CC.	DTW$^{glob.}$	Accuracy
Configuration	Ground truth	700 ± 20	0.10 ± 0.02	0	100
	DLS	529 ± 41	0.17 ± 0.02	129 ± 16	**9 ± 1**
	SimHawNet	**832 ± 165**	**0.06 ± 0.02**	**100 ± 26**	**5 ± 1**
SBM-b	Ground truth	698 ± 8	0.25 ± 0.02	0	100
	DLS	499 ± 15	0.15 ± 0.01	152 ± 7	**15 ± 1**
	SimHawNet	**865 ± 200**	**0.20 ± 0.08**	**115 ± 19**	**7 ± 2**
MID	Ground truth	1272	0.13	0	100
	LSH	15532 ± 2253	0.46 ± 0.03	1220 ± 195	0
	DLS	<u>698 ± 35</u>	**0.18 ± 0.03**	<u>223 ± 24</u>	**45 ± 1**
	SimHawNet	**1074 ± 37**	<u>0.29 ± 0.02</u>	**155 ± 12**	<u>20 ± 1</u>

→ SimHawNet can generate structurally accurate results, as shown by our results on CC, number of events and accuracy in synthetic and real datasets.

Results



(a) Config



(b) Enron

→ SimHawNet adapts to the temporal structure of diverse dynamics

Take Home Message

→ Hawkes process framework for edge-specific inter-arrival times

→ **SimHawNet:** simulation based on the learned model using Adapted Ogata's thinning algorithm

→ Evaluation framework on dynamic synthetic networks derived from standard static simulations

→ Relevant performance metrics

Open Source code